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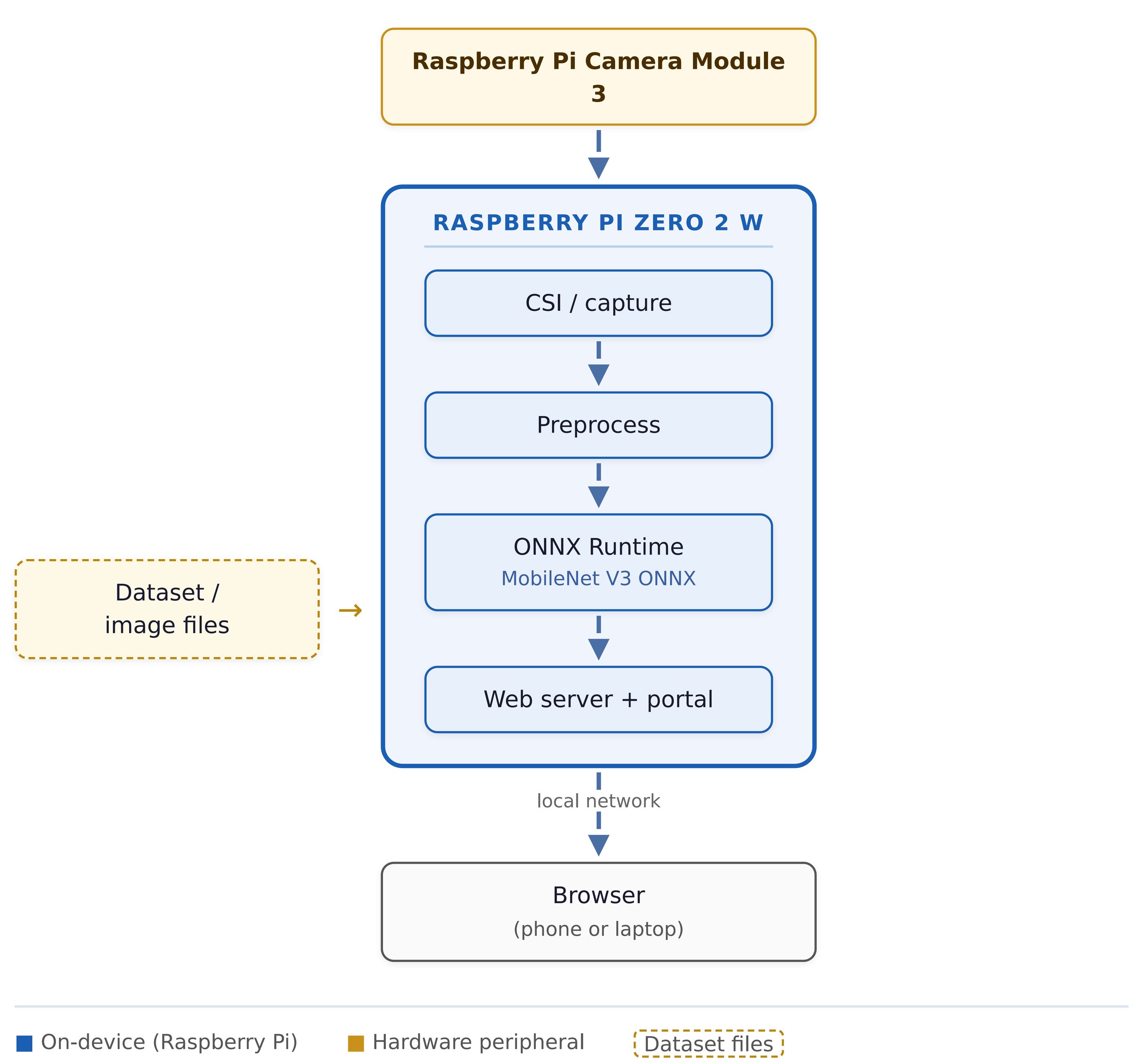
Edge Deployment: Plant Health Classification with MobileNet V3 on a Raspberry Pi Zero 2 W

The Problem

Pests and diseases destroy up to **40% of global crops** every year, about **US $220 billion** in losses. Early, on-site detection from leaf images, protects yield and cuts blanket pesticide use.

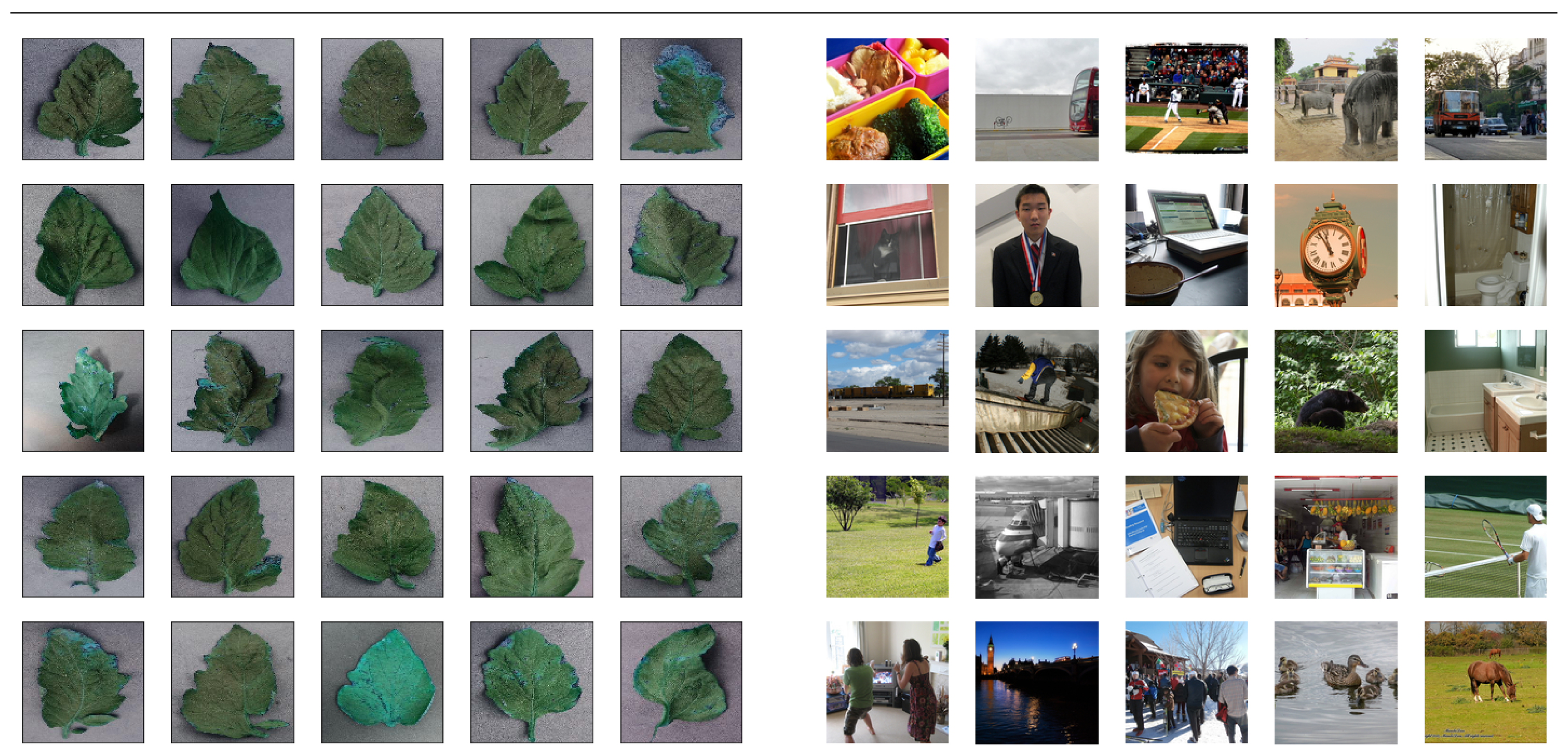
Goal: a self-contained classifier that labels a live camera frame as **Healthy**, **Diseased**, or **Background** — running entirely on a $15, 512 MB Raspberry Pi Zero 2 W with *no* cloud, GPU, or NPU.

System Architecture



Camera (or stored images) → on-device preprocess → ONNX Runtime → web portal streamed over the LAN.

Dataset — 3 classes, 12,606 images

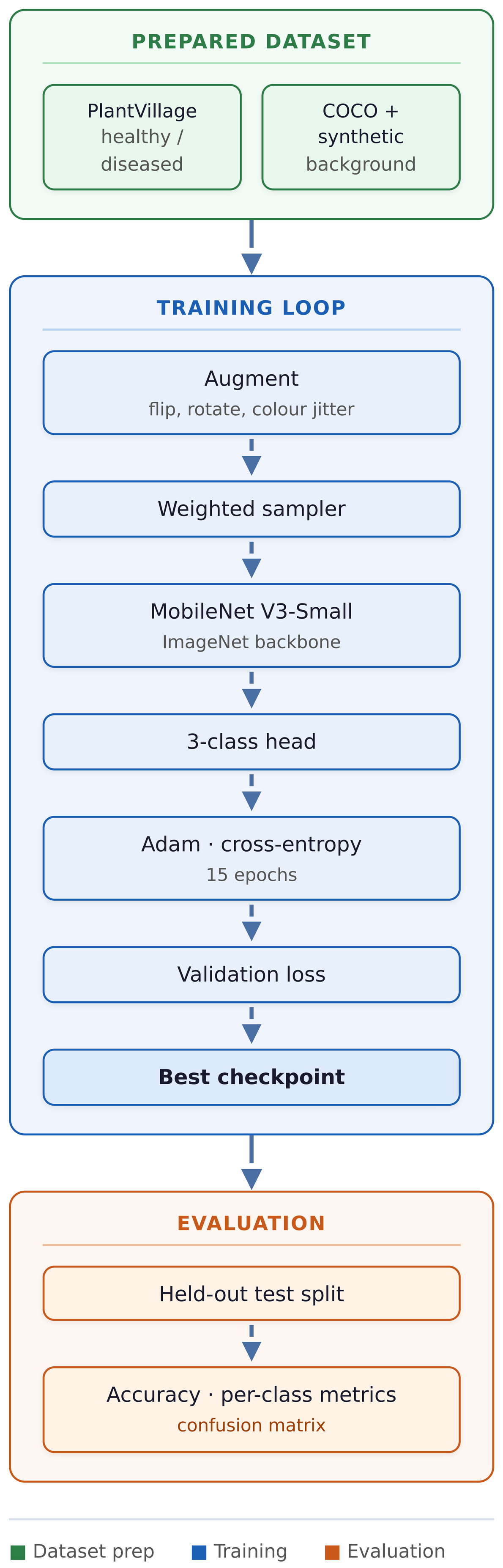


PlantVillage leaves collapsed to healthy / diseased; **COCO 2017** + field negatives form the background class that lets the model reject non-leaf scenes. Split **70 / 15 / 15**.

Key Design Choices

- **3-class topology** — a built-in *background* class rejects empty scenes with no extra detection model.
- **FP32 over INT8** — the Cortex-A53 lacks `SDOT`/`UDOT`, so INT8 falls back to slow software; FP32 uses mature NEON FMA instructions.
- **Native C++ runtime** — keeps memory far under the 512 MB ceiling; zero-allocation hot loop.
- **MobileNet V3-Small** — only **1.52 M** parameters.

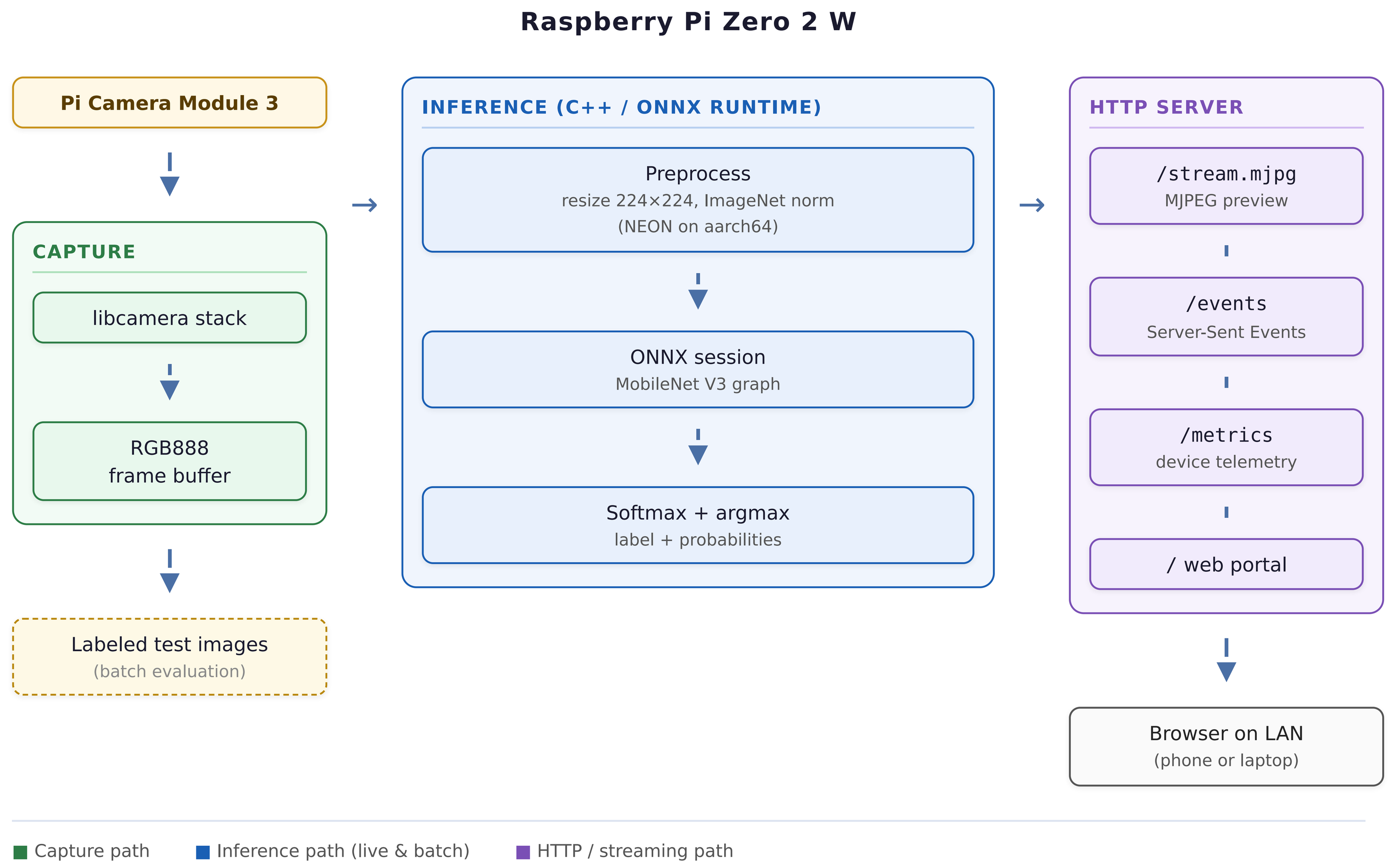
Training Pipeline



Live Results

68	31	8
ms inference (live)	ms encoding (live)	images / sec

On-Device Runtime (C++)

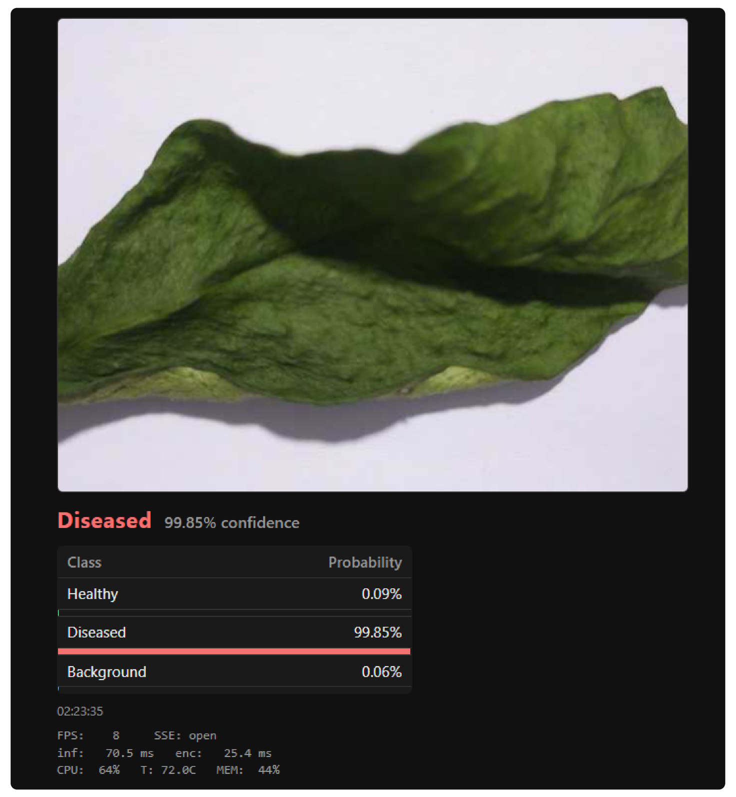


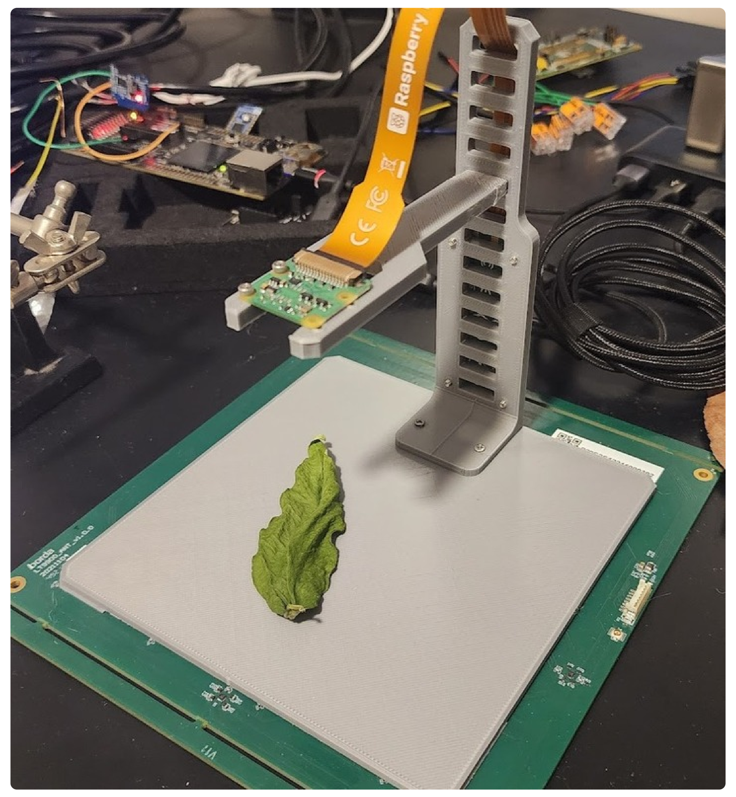
Edge Benchmarks

Build	Split	Acc.	img/s
FP32 (4 thr.)	3-class	**99.81%**	**15.73**
FP32 (1 thr.)	2-class	99.83%	8.03
INT8 (4 thr.)	3-class	97.05%	9.83
INT8 (1 thr.)	3-class	97.21%	6.04

Per-class F1: Healthy 0.995, Diseased 0.998, Background 0.999. INT8 is *slower* here — wrong tool for this silicon. These benchmarks does not include the encoding, and any other overhead. And just to showcase the performance of the ONNX runtime model with different threading and quantization.

Live Demo & Test Bench





Web portal shows live prediction, per-class confidence, FPS, and CPU temperature. A 3D-printed PLA bench fixes camera height and lighting for repeatable physical validation.

Conclusion

A native C++ end to end pipeline runs MobileNet V3 at ~**8 img/s** on a Pi Zero 2 W (live demo: **inference latency** ~**68 ms**, **encoding latency** ~**31 ms** per frame). The background class removes the need for a second model, and FP32 beats INT8 on this CPU. *Future work:* fine-tune on deployment-camera captures and add field validation in live greenhouses.